SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	SUNDARESAN.R.U	**Reg. No.:**	192424359
Section / Batch:	2025	**Date:**	10/07/2026

Code of Conduct

I__SUNDARESAN____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student____SUNDARESAN_____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 -6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
	GRAND TOTAL:	**100 Marks**	

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QUESTIONS

- **Q1** C Code – Array Traversal 10 marks
- **Q2** C Code – Linear Search 10 marks
- **Q3** C Code – Binary Search Mid 10 marks
- **Q4** C Code – Pointer with Array 10 marks
- **Q5** C Code – Array Update 10 marks
- **Q6** C Code – Linked List Node 10 marks

Q1 **C Code – Array Traversal** 10 marks

```
#include <stdio.h>
int main() {
int arr[] = {5, 10, 15, 20};
int i, sum = 0;
for(i = 0; i < 4; i++)
sum += arr[i];
printf("%d", sum);
return 0;
}
```

Your Answer

50

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QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code - Linked List Node
10 marks

Q7

Q2 **C Code – Linear Search** 10 marks

```
#include <stdio.h>
int main() {
int arr[] = {2,4,6,8,10};
int key = 8;
int i;
for(i=0;i<5;i++){
if(arr[i]==key){
printf("%d",i);
break;
}}}
```

Your Answer

3

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QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code – Linked List Node
10 marks

Q3 **C Code – Binary Search Mid**

10 marks

```
#include<stdio.h>
int main(){
int low=0,high=7;
int mid;
mid=(low+high)/2;
printf("%d",mid);
}
```

Your Answer

3

Save Answer

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QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code – Linked List Node
10 marks

Q8 **C Code - Nested Loop**

10 marks

```
for(i=0;i<n;i++)
for(j=0;j<n;j++)
printf("*");
```

Your Answer

$O(n^2)$

Save Answer

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QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code - Linked List Node
10 marks

Q9 **C Code - Binary Search Complexity**

10 marks

```
while(low<=high){
mid=(low+high)/2;
}
```

Your Answer

O(log n)

Save Answer

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QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code – Linked List Node
10 marks

Q10 **C Code - Pointer Increment**

10 marks

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

Your Answer

10

Save Answer